

Project Design Phase Proposed Solution Template

Date	13 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Manually evaluating a student's spoken explanation of a concept does not scale: instructors have limited time to listen to and grade every audio submission consistently, and students have no way to objectively check their own understanding before a graded assessment. This leads to slow, inconsistent feedback for students and a heavy, repetitive grading workload for instructors, particularly for oral or verbal assessment formats.
2.	Idea / Solution description	The Voice-Based Concept Understanding Analyzer (VBCUA) is a Streamlit web application that automatically assesses a student's spoken explanation of a concept. The workflow is: • The student uploads a recorded audio response (WAV, MP3, or M4A). • OpenAI Whisper transcribes the audio to text. • Sentence Transformers (all-MiniLM-L6-v2) generates embeddings for the transcript and a stored reference answer, and computes their cosine similarity. • The similarity score is converted into a percentage, a grade tier (Excellent / Good / Fair / Needs Improvement), and descriptive feedback. • A downloadable PDF report is generated containing the transcript, score, grade, and feedback.
3.	Novelty / Uniqueness	Unlike generic transcription tools, VBCUA does not just convert speech to text — it evaluates that text against a reference answer to produce an objective concept-understanding score. Unlike text-based quizzes or multiple-choice auto-graders, VBCUA assesses a student's own spoken articulation of a concept, which better reflects genuine understanding than selecting a pre-written answer. It combines automatic speech recognition and semantic similarity analysis into a single, lightweight, purpose-built pipeline for educational spoken-response evaluation, rather than requiring separate manual tools stitched together.
4.	Social Impact / Customer Satisfaction	Reduces the repetitive grading workload on instructors, freeing time for more meaningful, higher-value teaching interactions.

S.No.	Parameter	Description
		Gives students immediate, objective feedback on their spoken explanations, supporting self-paced practice and reducing uncertainty and anxiety before formal oral assessments. Encourages genuine conceptual understanding over rote memorisation, since the assessment is based on the student's own explanation rather than a fixed-choice answer. Provides a documented, downloadable record of performance that students can use to track their own preparation.
5.	Business Model (Revenue Model)	The current version of VBCUA is an academic prototype and does not implement any billing, licensing, or monetisation logic. The following outlines a plausible revenue model should the solution be taken beyond the current prototype stage: • Freemium model: free access for individual students and small classes, with a paid tier for institutions requiring features such as multiple reference answers, batch processing, or performance dashboards. • Institutional licensing: a per-institution or per-department subscription for schools, colleges, and coaching centres to deploy VBCUA for oral assessments at scale. • LMS integration / API licensing: offering VBCUA's transcription-and-scoring pipeline as an add-on or API integration for existing learning management systems (LMS).
6.	Scalability of the Solution	The current implementation runs as a single local Streamlit process with in-memory model caching and Streamlit session state, and is designed for single-user, single-session use. The underlying pipeline is modular (separate transcription, semantic analysis, scoring, and reporting components), which makes it straightforward to scale in future by: • Deploying the application to a cloud platform with a persistent, database-backed session store to support many concurrent users. • Introducing a job queue for transcription and analysis so multiple submissions can be processed in parallel. • Supporting multiple reference answers and subjects so the tool can scale across courses and departments, not just a single concept. These scalability enhancements are identified as future scope and are not part of the current implementation.